HARDWARE × MODELS × REAL-WORLD SPEED

# Intelligence. On your desk.

A field guide to capable local coding and tool-using agents.
14 hardware configurations. Seven local models + four Claude Opus reference points.

**Model capability, memory, speed and cost.**
Reported benchmarks and clearly labelled estimates for local AI hardware.

## 01 / The hardware-to-model map

M reported measurement  E estimate

**35 tok/s gate:** choose the highest-scoring fitting model whose decode range reaches ≥35 tok/s, then sort by score and speed ↑. One active request; a qualifying upper bound is not a guaranteed minimum.

| DEVICE / CONFIGURATION | APPROX. USD | MEMORY INSTALLED | BANDWIDTH GB/S | AI COMPUTE DENSE PEAK | BEST-FIT SHORTLIST MODEL / PRECISION | DECODE TOKENS / SEC |
|---|---|---|---|---|---|---|
| **MacBook Pro · M5** [H1]<br>10-core GPU | **$2.5k**<br>System | **32 GB**<br>Unified | 153 | **~16 TFLOPS** FP16 GPU · E | Qwen3.6-35B-A3B · 4-bit MLX<br>oMLX 0.3.6 · ~20 GB peak | **51–60** M<br>1K–4K · two runs · B12 |
| **MacBook Pro · M5 Pro** [H1]<br>20-core GPU | **$3.7k**<br>System | **64 GB**<br>Unified | 307 | **~36 TFLOPS** FP16 GPU · E | Qwen3.8-27B · dynamic Q4<br>MTPLX · native MTP drafting | **21–44** E<br>Upper end: repetitive editing |
| **GeForce RTX 4090** [H3] | **$2–2.5k**<br>Card est. | **24 GB** | 1,008 | **661 TFLOPS** FP8 | Qwen3.8-27B · Q4_K_S<br>llama.cpp · no MTP | **38–46** M<br>64K → 4K · B10 |
| **Radeon AI PRO R9700** [H4] | **$1.7k**<br>Card | **32 GB** | 640 | **383 TFLOPS** FP8 | Qwen3.8-27B · Q4-class + MTP<br>llama.cpp Vulkan · MTP=2 | **51.8** M<br>AMD vendor · preliminary · B8 |
| **GeForce RTX 5090** [H5] | **$6.4–7.4k**<br>Card est. | **32 GB** | 1,792 | **1,676 TFLOPS** FP4 | Qwen3.8-27B · Q4_K_M<br>LM Studio / CUDA | **59** M<br>6.1K · no MTP · B3 |
| **Ryzen AI Max+ 395 / Halo** [H2]<br>Radeon 8060S | **$2–4k**<br>System est. | **128 GB**<br>GPU share varies | 256<br>LPDDR5X-8000 | **110 T ops/s**<br>Mixed peak: GPU + NPU<br>60 FP16 + 50 INT8 | Qwen3.8-Flash-Next · UD-IQ4_XS<br>llama.cpp ROCm · MTP + n-gram drafting | **32–47** M<br>8K new code → rewrite · B7 |
| **1 × DGX Spark** [H7]<br>Single NVIDIA GB10 system | **$4.7k**<br>System | **128 GB** | 273 | **500 TFLOPS** FP4 | Qwen3.8-Flash-Next · NVFP4<br>vLLM + MTP3 · table on SSD | **37–46** M<br>Prose → code · B5 |
| **MacBook Pro · M5 Max** [H1]<br>40-core GPU | **$6k**<br>System est. | **128 GB**<br>Unified | 614 | **~70 TFLOPS** FP16 GPU · E | Qwen3.8-Flash-Next · dynamic Q4<br>MTPLX 2.12 + MTP · n-gram table on SSD | **63.5** M<br>65.5K prompt · B1 |
| **RTX PRO 6000 Blackwell** [H6]<br>Workstation Edition | **$16–20k**<br>Card | **96 GB** | 1,792 | **2,015 TFLOPS** FP4 | Qwen3.8-Flash-Next · NVFP4<br>vLLM · n-gram table in ~100 GB host RAM | **81–88** M<br>No MTP; 120–173 MTP · B9 |
| **Mac Studio · M5 Ultra** [H8]<br>80-core GPU | **$12k**<br>System est. | **256 GB** | 1,200 | **~140 TFLOPS** FP16 GPU · E | GLM-5.3-Flash · oQ4e / mixed 4/8-bit<br>oMLX + DFlash2 drafter | **44–62** M<br>4K → short prompt · B17 |
| **2 × DGX Spark** [H7] | **$9.4k**<br>Systems | **256 GB**<br>128 GB per node | 2 × 273<br>Distributed | **1,000 TFLOPS** FP4 · 2 devices | GLM-5.3-Flash · EXL3 4-bit<br>TP2 · DFlash2 drafter · 200G RoCE | **47–63** M<br>Prose → code · B16 |
| **2 × RTX PRO 6000 Blackwell** [H6] | **$32–40k**<br>Cards est. | **192 GB**<br>96 GB per GPU | 2 × 1,792<br>Not shared | **4,030 TFLOPS** FP4 · 2 GPUs | GLM-5.3-Flash · IQ4_XS<br>Layer split · ~157 GB weights · fit projected | **41–70** E<br>Best ~55 · B13 anchor |
| **M5 Ultra · 512 GB** [H8]<br>Ships late October | **$15–18k**<br>System est. | **512 GB** | 1,200 | **~140 TFLOPS** FP16 GPU · E | GLM-5.3 · IQ4_XS / oQ4<br>Needs DFlash2 drafter · ~365–420 GB weights | **18–40** E<br>Drafting · 15–25 without |
| **DGX Station · GB300** [H9]<br>OEM order · deskside | **$99–189k**<br>System | **748 GB**<br>252 HBM + 496 | 7,100 / 396<br>HBM / LPDDR5X | **15,000 TFLOPS** FP4 | GLM-5.3 · IQ4_XS, split HBM / CPU<br>Needs DFlash2 drafter + split-memory backend | **19–40** E<br>Drafting · 16–25 without |

**Compute:** Dense peaks only; no sparsity multiplier. GPU figures are TFLOPS at the stated precision. Macs exclude the separately implemented Neural Engine: its dense rating is unverified; GPU FP16 and INT8 peaks cannot be added. Halo sums 60 FP16 TFLOPS (GPU) + 50 INT8 TOPS (NPU): a mixed-precision theoretical aggregate for separate concurrent engines, not measured LLM throughput. Shared power/memory can limit simultaneous peaks. **Prices:** approximate USD; cards exclude host. **M / E:** third-party measurement / forecast. 128K memory budgets exclude unquantified DFlash additions. Two PRO Server cards separately report **164.8 tok/s** with another quantization (B19).

## 02 / Local models & the frontier trajectory

BASELINE Claude Opus · cloud yardstick, not runnable locally  AA Intelligence Index v4.3.2 · higher is better

**10 independent evaluations:** agents 30% · coding 20% · scientific reasoning 20% · general 30%. **Opus trajectory:** 4.6 (Feb) 32* → 4.8 (May) 42 → 5 (Jul) 51 → 5.5 (Sep) 58. The best local-fitting model, GLM-5.3 (45), sits between May's Opus 4.8 and July's Opus 5.

| MODEL / TESTED SETTING | AA SCORE | PARAMETERS / TYPE | MEMORY ESTIMATE · 128K CONTEXT · 1 ACTIVE REQUEST |
|---|---|---|---|
| **Claude Opus 5.5** BASELINE<br>Max · cloud ceiling<br>Released 22 Sep 2026 | 58 | **Undisclosed**<br>Architecture undisclosed | **Cloud only · weights and serving memory undisclosed** |
| **Claude Opus 5** BASELINE<br>Max · cloud reference<br>Released 24 Jul 2026 | 51 | **Undisclosed**<br>Architecture undisclosed | **Cloud only · frontier trajectory reference** |
| **GLM-5.3**<br>Max<br>Released 18 Aug 2026 · API notes | 45 | **~753B total · ~40B active**<br>MoE | **365 GB weights + ~13 GB cache ≈ 378 GB resident**<br>IQ4_XS · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB |
| **Claude Opus 4.8** BASELINE<br>Max · cloud reference<br>Released 28 May 2026 | 42 | **Undisclosed**<br>Architecture undisclosed | **Cloud only · frontier trajectory reference** |
| **GLM-5.3-Flash**<br>Reasoning<br>Released 26 Aug 2026 | 42 | **320B total · 18B active**<br>MoE | **157–200 GB weights + ~2 GB cache ≈ 159–202 GB resident**<br>Q4-class · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB<br>Expanded-KV backends need ~95–105 GB cache instead. |
| **Qwen3.8-Flash-Next**<br>Reasoning<br>Released 26 Aug 2026 | 40 | **~180B total · 6B active**<br>MoE + lookup · 125B backbone | **~83 GB weights + ~4 GB cache ≈ 87 GB resident**<br>MTPLX dynamic Q4 · 16-bit cache · +32 GB lookup tables on SSD · runtime: +4–8 GB |
| **DeepSeek V4.1 Flash**<br>Max<br>Released 10 Sep 2026 | 39 | **~763B total · 16B active**<br>MoE + Engram · 552B backbone · decode | **~316 GB weights + ~0.2–0.6 GB cache ≈ 317 GB resident**<br>DwarfStar Q4 · packed FP4–BF16 cache scenarios · +203 GB Engram on SSD · runtime: +8–16 GB |
| **Qwen3.8-27B**<br>Xhigh<br>Released 14 Aug 2026 | 34 | **27B total · 27B active**<br>Dense | **20.4 GB weights + ~9.3 GB cache ≈ 30 GB resident**<br>MTPLX dynamic Q4 · 16-bit cache · SSD tables: none · runtime: +4–8 GB |
| **DeepSeek V4 Flash / 0731**<br>Max · text model<br>Released 31 Jul 2026 · checkpoint | 34 | **284B total · 13B active**<br>MoE | **~155 GB weights + ~1 GB cache ≈ 156 GB resident**<br>Q4_K_XL · compressed attention, BF16 cache · SSD tables: none · runtime: +8–16 GB |
| **Claude Opus 4.6** BASELINE<br>Max · AA estimate<br>Released 5 Feb 2026 | 32 | **Undisclosed**<br>Architecture undisclosed | **Cloud only · frontier trajectory reference**<br>AA marks 32 as estimated; independent evaluation forthcoming. |
| **Qwen3.6-35B-A3B**<br>Reasoning · fallback<br>Released 16 Apr 2026 | 18 | **35B total · ~3B active**<br>MoE | **~22 GB weights + ~3 GB cache ≈ 25 GB resident**<br>Q4 · 16-bit cache · runtime: +4–8 GB |